

ROB5014 Mathematics and Simulation for Robotics

HW1

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1. Simulate the following simple harmonic motion of a simple mass-spring system without friction with 4 methods (EEI, IEL, SEI, PMI).
mass $m = 0.01\text{kg}$, spring constant $k = 100\text{N/m}$, No external force $f = 0$, Initial condition $x_0 = 0.1\text{mm}$
You need to show the position of the mass as a function of a time in a graph where 4 results are easy to compare. You need to submit the result as well as the source codes.
 2. Simulate the double-pendulum motion using PGS algorithm in your favorite programming language. Use the following physical dimension.
Mass $m_1 = m_2 = 1\text{kg}$, length $l_1 = l_2 = 100\text{mm}$, (assume thin rod)
If you want, you can further assume the model. In this case, you must describe what your assumptions are.
You need to report the result as well as the source codes.